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The nano-mechanical morphology of shale

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Abstract

Shale, the sealing formations in most hydrocarbon reservoirs, is made of highly compacted clay particles of sub-micrometer size, nanometric porosity and different mineralogy. In this paper, we propose and validate a technique to identify the nano-mechanical morphology of such a nanocomposite material. In particular, by means of a massive nanoindentation campaign at two different scales on a large range of shale materials, we show that the highly compacted plate- or sheet-like clay particles have a distinct nano-mechanical morphology with no privileged orientation of the particle-to-particle contact surface, as evidenced by a mechanical percolation threshold of $\eta_0 \geqslant 0.5$. Furthermore, the nanoindentation results provide strong evidence that the nano-mechanical elementary building block of shales is transversely isotropic in stiffness, and isotropic and frictionless in strength. These observations lead to a sphere-like mechanical morphology for visibly plate- or sheet-like clay particles. The contact forces between the sphere-like particles activate the intrinsicly anisotropic elastic properties within the clay particles and the cohesive bonds between the clay particles. The mechanical stiffness and strength properties of porous clay scale with the clay packing density toward a unique set of shale-invariant material properties. The determination of mechanical microstructure and invariant material properties are of great importance for the development of predictive microporomechanical models of the stiffness and strength properties of shale. The approach presented here also applies to other chemically and mechanically complex materials exhibiting nanogranular behavior.
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Keywords: Shale; Clay; Nanoindentation; Granular material; Packing density; Morphology

1. Introduction

Shale is a multi-phase, multi-scale, transversely isotropic, and compositionally diverse sedimentary rock (Fig. 1). It is composed primarily of sedimented clay particles, and some quantities of larger, silt-sized inclusions, and is especially critical for the oil industry in both exploration and exploitation applications (Thomsen, 2001). The complex nature of shale materials, however, has challenged researchers to understand how shales behave mechanically, and how their material properties may be predicted (see, for instance, Kaarsberg, 1959; Hornby et al., 1994; Hornby, 1998; Johnston and Christensen, 1995; Sayers, 1994, 1999; Vernik and Nur, 1992, among many others). The widely documented diversity of shale

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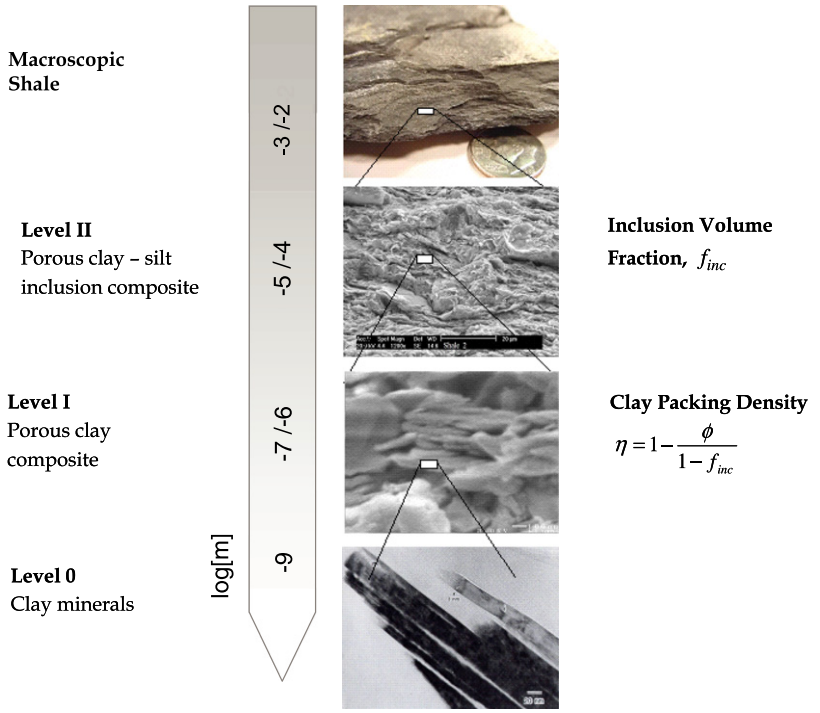


Fig. 1. Multiscale model for shale, adapted from Ulm et al. (2005). The considered scales and their associated volume parameters are shown. The Level II and Level I images come from ESEM imaging. The width of the Level II image is about 75 μm. The width of the Level I image is about 1.75 μm. The Level 0 image is a TEM image from Lee et al. (1991). The width of the Level 0 image is 325 nm.

materials currently demands detailed localized testing in engineering applications, motivating the need for improved understanding and predictive models.

Development of micromechanical models that could predict macroscopic material properties depends on an understanding of the microstructure of the material. Researchers studying shale thus far have focused their efforts to model anisotropy of shale by considering particle shapes that reflect visually observed microstructures. The plate- or sheet-like structure commonly attributed to clay minerals (see Fig. 1; Level 0) has been captured by models describing the orientation of these solid particles (Sayers, 1994; Hornby et al., 1994; Draege et al., 2006) or by models describing the orientation of the pore spaces between particles (Ulm et al., 2005). These models have been successfully applied in limited cases, but their required calibration of orientation distribution functions or parameters is a challenge, especially given the diversity of shale materials.

Is such a literal translation of visual structure to mechanical morphology the best approach? While visual observations are useful, careful analysis of the mechanical responses of materials may be more enlightening for development of multi-scale micromechanical models. With an understanding of shale as a multi-phase, multi-scale material (Ulm and Abousleiman, 2006; Ulm et al., 2005), the nanoscale mechanical behavior of shale materials is of particular interest. Continuing innovation in instrumented indentation and analysis permits the measurement of mechanical behavior at the nanoscale. From analysis of these measurements, we ask, what is the nano-mechanical morphology? Does this nano-mechanical morphology help lead to the determination of material invariant shale properties at the nano-scale?

We demonstrate here, via the experimental techniques of instrumented micro and nanoindentation, that there is a mechanical morphology at the nanoscale which is distinct from the microscopically visible structure of shale materials. As a consequence, it is possible to identify shale-invariant material properties for shale, i.e. material properties that are not affected by the specific mineralogy or porosity of a specific shale material. The goal of this paper, therefore, is to introduce a new understanding of the mechanical behavior of shale materials from the nano- to the macro-scale.

2. Materials and methods

2.1. Nanoindentation on natural composites

While the use of instrumented nanoindentation has been extensively studied in the past two decades (for recent reviews, see Cheng and Cheng, 2004; Oliver and Pharr, 2004), the tools have been limited, until recently, to either homogeneous samples or to layered samples (thin films) whose thicknesses are previously known (see e.g. Perriot and Barthel, 2004). Recently developed and refined techniques, however, have extended the application of nanoindentation to heterogeneous, composite materials (Ulm et al., 2003, 2007; Constantinides et al., 2006; Constantinides and Ulm, 2007).

Briefly, we recall that nanoindentation consists of making contact between a sample and an indenter tip of known geometry and mechanical properties, followed by a continuously applied and recorded change in load, P, and depth, h. Typical tests consist of a constantly increasing load, followed by a short hold and then a constant unloading; a P–h curve is reported. The analysis of the P–h curve proceeds by applying a continuum scale model to derive an indentation modulus, M, and a hardness, H:

$$M = \frac{\sqrt{\pi}}{2}\frac{S}{\sqrt{A_c}}, \tag{1}$$

$$H = \frac{P}{A_c}, \tag{2}$$

where $S = \frac{dP}{dh}$ is the (measured) initial slope of the unloading branch of the P–h curve, P is the (measured) maximum indentation load, and A_c is the projected contact area of the indenter on the sample surface. The projected contact area, A_c, is typically determined as a function of the (measured) maximum indentation depth, h_{max} (Oliver and Pharr, 2004).

Recognizing the high heterogeneity of shale at the nano- and micro-scale, applying the indentation technique is a challenge, as it is difficult to choose to indent on a specific material phase with sufficient repeatability. To address this challenge, Ulm and co-workers (Ulm et al., 2003; DeJong and Ulm, 2007; Constantinides et al., 2006; Constantinides and Ulm, 2007) realized the advantage of performing large grids of indentations on heterogeneous samples. They proposed that if the grid size and indentation depth are chosen properly, each indentation test may be treated as an independent statistical event. Then statistical techniques may be applied to the results.

2.1.1. Scale separability

The grid indentation technique introduces new length scales to the problem. These length scales enter as criteria for statistical independence and statistical sampling, as well as the scale separability condition from continuum mechanics. Considering statistical independence first, we must choose a grid size, l, which is much larger than the imprint of the indentation (a function of the indentation depth, h). To avoid sampling effects, a large number of tests, N, should be carried out over an area much larger than the length scale of the individual material phases, D. On the other hand, if we wish to obtain properties of individual material phases, the indentation depth, h, should be much smaller than the length scale of the phases, D. In addition, finite element simulations have shown that a Berkovich indentation test measures the elastic response of a material volume 3–5 times the depth of indentation, h (Larsson et al., 1996) adding to the requirement that h should be much smaller than D. These conditions may be summarized:

$$3h \ll D \ll l\sqrt{N}. \tag{3}$$

Moreover, because the analysis and interpretation of the indentation tests will depend on continuum mechanics analysis, there is an additional challenge of respecting the separation of scale condition:

$$d \ll \mathscr{L} \ll h, \tag{4}$$

where $\mathscr{L}$ is the characteristic size of the representative elementary volume (rev), which must be much greater than the size of the largest heterogeneity of size d contained in the rev and much smaller than the indentation depth, h, which defines the volume of material sensed in an indentation test.

An additional experimental challenge in indenting on natural materials is to polish the sample to provide a smooth and flat surface to indent on, while minimizing the sample disturbance. The Hertz contact problem employed in analysis of nanoindentation assumes a perfectly flat infinite half space. A proper polishing procedure should provide a reasonable approximation of this assumption. Researchers have shown that the indentation depth should be larger than the root mean squared

(RMS) roughness, R_q, although how much larger is still a somewhat open question: Bobji and Biswas (1998) suggest, as a result of simulations and experiments on copper with rounded conical indenters, that the indentation depth should be at least an order of magnitude greater than the RMS roughness. Donnelly et al. (2006), while performing Berkovich nanoindentation experiments on bones, have demonstrated that indentation depths greater than 3 times the RMS roughness are sufficient to avoid effects of roughness.This condition is summarized

$$R_q < 3h. \tag{5}$$

These three conditions on indentation depth, h, and the condition on the grid size, l, and number of tests, N, are necessary for a rigorous application of the grid indentation technique.

2.1.2. Deconvolution technique

The deconvolution technique allows the estimation of mechanical phase properties from a grid indentation data set with the results of many individual indentation tests, given that the scale separability conditions (Eqs. (3)–(5)) have been satisfied. Each indentation test is considered as a single statistical event, and the extracted indentation modulus, M, and hardness, H, are considered as random variables. The aim of the deconvolution technique is to identify, from the experimental distributions of M and H, the number of mechanically active material phases; and the mechanical phase properties. It is analytically more convenient to deconvolute the cumulative distribution functions (CDF) than to deconvolute the probability density functions (PDF), because generation of the experimental PDF requires a choice of bin-sizes. The PDFs are more physically intuitive, however, so for visualization of the results we will return to PDFs.

The deconvolution technique begins with the generation of the experimental CDF. Let N be the number of indentation tests performed on a specimen, and $\{M_i\}$ and $\{H_i\}$ $(i = 1, N)$ the sorted values of the measured indentation modulus M and indentation hardness H. The N points of the experimental CDF of M and H, denoted by F_M and F_H respectively, are obtained from:

$$\left.\begin{aligned} F_M(M_i) &= \tfrac{i}{N} - \tfrac{1}{2N} \\ F_H(H_i) &= \tfrac{i}{N} - \tfrac{1}{2N} \end{aligned}\right\} \quad \text{for } i \in [1; N]. \tag{6}$$

Once the experimental CDFs are known, the form of the model CDF's are specified. Consider the heterogeneous material to be composed of $j = 1, n$ material phases with sufficient contrast in mechanical phase properties. Each phase occupies a surface fraction, f_j, of the indented surface. The distribution of the mechanical properties of each phase is assumed to be correctly approximated by Gaussian distributions, identified by the mean values μ_j^M and μ_j^H and the standard deviations s_j^M and s_j^H, of the indentation modulus M and the indentation hardness H, respectively. The CDF for each (Gaussian distributed) phase is given by

$$\begin{aligned} &j = 1, n; \\ &F(X_i; \mu_j^X, s_j^X) = \frac{1}{s_j^X\sqrt{2\pi}} \int_{-\infty}^{X_i} \exp\left(\frac{-(u-\mu_j^X)^2}{2(s_j^X)^2}\right) \mathrm{d}u; \\ &X = (M, H). \end{aligned} \tag{7}$$

The $n \times 5$ unknowns $\{f_j, \mu_j^M, s_j^M, \mu_j^H, s_j^H\}$, $j = 1, n$ are determined by minimizing the difference between the experimental CDFs and the weighted model-phase CDFs:

$$\begin{aligned} \min \Bigg[&\sum_{i=1}^{N} \left(\sum_{j=1}^{n} f_j F(M_i; \mu_j^M, s_j^M) - F_M(M_i) \right)^2 \\ &+ \sum_{i=1}^{N} \left(\sum_{j=1}^{n} f_j F(H_i; \mu_j^H, s_j^H) - F_H(H_i) \right)^2 \Bigg] \end{aligned} \tag{8}$$

s.t.

$$\sum_{j=1}^{n} f_j = 1,$$

where the constraint of the minimization problem requires that the surface fractions of the different phases sum to one. To ensure that phases have sufficient contrast in properties, and thus to avoid that two neighboring Gaussians do not overlap, the optimization problem is additionally constrained by DeJong and Ulm (2007):

$$\mu_J^X + s_J^X \leqslant \mu_{J+1}^X - s_{J+1}^X, X = (M, H). \tag{9}$$

The deconvolution is performed by a non-linear least-squares solver in Matlab. The results of the deconvolution technique are estimates of the mean

and standard deviation of indentation modulus and hardness for each mechanical phase.

2.2. Tested materials

Seven different shale materials, obtained from core samples, were tested in this study. Core sections were further drilled or trimmed to expose surfaces normal to observed bedding planes and parallel to observed bedding planes. Information regarding the geological origin of these materials was generally not provided.

2.2.1. Multiscale model of shale

Previous research has demonstrated that shale is a transversely isotropic, multi-phase, multi-scale material (Ulm et al., 2005). The transversely isotropic behavior is associated with the observed bedding planes of shale materials. We denote indentations performed normal to the plane of isotropy as the $x3$-direction and indentations performed parallel to the axis of symmetry as the $x1$-direction. The multiscale model is shown in Fig. 1. The macroscopically observed shale, at a minimum length scale of centimeters, is seen at a scale below as a porous clay (which has been lithified over time) and silt inclusion (such as grains of Quartz or Feldspar) composite (Fig. 1; Level II). In turn, this lithified porous clay is seen at another scale below as a composite of lithified clay minerals (which may possess some structural water) and pores filled with some bulk fluid (Fig. 1; Level I). The binding material of shales is the porous clay. The mechanical properties of this porous clay are critical in explaining the mechanical properties of shales.

2.2.2. Mineralogy and volumetric parameters

Information regarding the porosity, bulk density, and mineralogy (in terms of a ratio of non-clay inclusion to solid volume fraction) was provided with the samples, and is summarized in Table 1. Preliminary investigations on shale have indicated that the clay packing density of the material is of critical importance to its mechanical behavior (Ulm and Abousleiman, 2006) (similar results were found in investigations on other silicate materials such as cements (Constantinides and Ulm, 2007)). For shales, this packing density neatly summarizes the porosity and mineralogy information about the material into one simply understood parameter.

Given the appropriate data, clay packing density is easy to calculate. The total volume is partioned into three categories, porosity, ϕ, clay volume fraction, f_{inc}, and non-clay inclusion volume, f_{inc}, that

Table 1
Properties of the studied shale samples

Sample		1	2	3	7	Light	Dark	Pierre	Low-clay
Mineralogy[a]									
$f^{\text{inc}}_{\text{min}}$		0.155	0.225	0.275	0.269	0.615	0.588	0.466	–
$f^{\text{inc}}_{\text{max}}$		0.164	0.257	0.276	0.279	0.643	0.620	0.524	0.560
f^{clay}		0.576	0.632	0.650	0.646	0.276	0.344	0.323	0.150
	Kao.	0.271	0.189	0.084	0.358	0.276	0.192	0.000	–
	I./S.	0.289	0.368	0.506	0.203	0.000	0.098	0.229	–
	Other	0.016	0.076	0.060	0.086	0.000	0.053	0.095	–
Density[b]									
ρ		2.20	2.43	2.55	2.51	2.48	2.57	2.25–2.4[c]	–
Porosity									
ϕ_{min}		0.260	0.133	0.075	0.075	0.065	0.037	0.168	0.290
ϕ_{max}		0.299	0.169	0.076	0.108	0.098	0.087	0.257	–
Porous clay Packing density									
η_{min}		0.647	0.782	0.896	0.853	0.746	0.790	0.517	–
η_{max}		0.689	0.827	0.897	0.897	0.829	0.904	0.647	0.341

Minimum volume fractions were calculated using mineralogy and bulk density information. Maximum volume fractions were calculated using mineralogy and MIP porosity information.

[a] Mineralogy in volume fractions: inc. = inclusion, Kaol. = kaolinite, I./S. = Illite/Smectite. Other = chlorite, glauconite, biotite.

[b] Density in grams per cubic centimeter.

[c] From Savage and Braddock (1991), this variation in bulk density gives the max and min packing density.

sum to one. This information permits the calculation of the clay packing density, η, given by

$$\eta = 1 - \frac{\phi}{1 - f_{\text{inc}}}. \tag{10}$$

The clay packing density may vary depending on how the porosity is measured. It is well documented that mercury intrusion porosimetry underestimates the actual porosity (Diamond, 2000), while estimation of porosity by using the bulk density and mineralogy information gives a higher value of porosity. This leads to a variation in the calculation of f_{inc} and η as values calculated from porosity are smaller than those calculated from the bulk density. The results of these calculations are also shown in Table 1.

The packing density contains information about porosity and mineralogy of shale samples. The tested shales have a relatively wide range of mineralogies and porosities, confirming the known diversity of shale materials.

2.2.3. Storage and surface preparation

The material samples (small core sections) were stored in conditions near to the natural relative humidity of the materials, either in desiccators with salt solutions or in tightly sealed vessels filled with decane. Samples were cut to expose surfaces parallel and perpendicular to the isotropy planes of the material.

As previously discussed, the analysis of nanoindentation assumes that indentation occurs on an infinitely flat surface. Although this is not possible in practice, a surface preparation program was developed which minimizes both the roughness and sample disturbance of the material. The samples are first mounted to stainless steel plates for mounting in the indenter. They are then ground by hand on 45 grit diamond paper (with the help of a jig) to make the surface flat and parallel with the mounting plate. The samples were then polished with a 1 μm diamond paste (oil-based) on TexMet pads (Buehler) mounted to a lapping wheel. From AFM testing, the RMS roughness obtained with this polishing procedure measurements was found to be on the order of 150–300 nm (Table 2).

Table 2
RMS Roughness of the polished shale samples, calculated over 20 μm by 20 μm areas (512 by 512 pixel resolution) measured with a non-contact tip

Shale	RMS roughness, R_q (nm)	
	$x1$-direction	$x3$-direction
1	–	–
2	256	204
3	128	152
7	223	227
Light	252	166
Dark	163	205
Pierre	246	329
Low-clay	–	–

Topography with a wavelength greater than 10 μm was filtered out.

3. Results

3.1. Choice of indentation depth

The proper choice of indentation depth is essential to satisfy the grid indentation scale separability conditions (Eqs. (3)–(5)) for the porous clay. We seek to test both the porous clay as an overall composite and as a locally packed state; we therefore require two sets of length scales to obtain two appropriate target indentation depths. Information obtained from environmental scanning electron microscope (ESEM) imaging (Fig. 1), mercury porosimetry investigations (Fig. 2), and Atomic Force Microscope (AFM) measurements (Table 2) helps identify the length scales in question.

First, consider the porous clay as a composite material. We seek first the largest heterogeneity, d, contained in an *rev* of porous clay. From porosime-

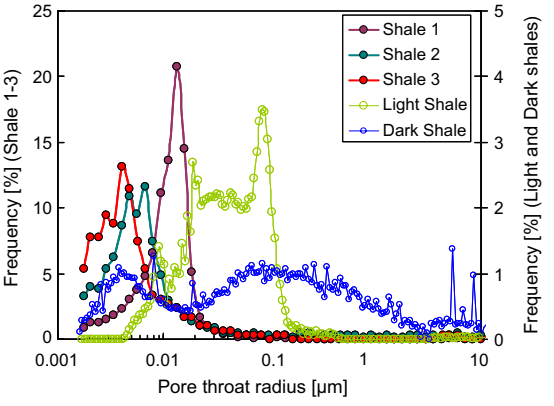


Fig. 2. Mercury Intrusion Porosimetry Results for Shales 1, 2, 3, Light, and Dark. Typical pore throat radii are on the order of 10–100 nm (data for shale 1–3 provided by Chevron; data for the light and dark shale provided by the Poromechanics Institute of the University of Oklahoma at Norman).

try (Fig. 2), we determine the largest heterogeneity, d, by identifying typical pore throat radii on the order of 10 nm. The RMS roughness measurements determined from the AFM tests (Table 2) are on the order of 150–300 nm. We wish to resolve the properties of the porous clay composite as distinct from the properties of the inclusion phases, with characteristic length scale, D. From the imaging experiments, we observe characteristic sizes of the inclusion (non-clay) material phases as small as 20 μm. Combining the lower limits, d and R_q, and the upper limit, D, we ideally wish to test with indentations that sense depths greater than 500 nm to less than 3 μm. Note that this range is significantly narrower than the ranges explored by other researchers in previous applications of the grid indentation technique for cement paste and its primary binding material, C–S–H (Constantinides and Ulm, 2007; DeJong and Ulm, 2007). Consequently, we must be especially careful to isolate the appropriate target indentation depth for sensing properties of the porous clay composite in shale materials. Finally, we need enough tests over a wide area to achieve statistical independence and good sampling representation. With about $N = 300$ tests, a grid size of $l = 30$ μm is more than sufficient.

Second, consider testing locally packed areas of the porous clay. Here we expect the largest heterogeneity, d, to correspond to the pore throat radii in the vicinity of a locally densely packed region of clay, so it is reasonable to assume that the pore throat radii values of just a few nanometers are applicable here. We may be limited by the roughness of the sample, but, as the least stringent of the length scale criteria, we relax this restriction for now. We return to the roughness issue in the discussion of these results. In the case of the locally densely packed region, we wish to resolve properties distinct from the porous clay composite, whose length scale, D, is identified above at around 1 μm. Consequently, we wish to test with indentation depths smaller than 300 nm (although depths larger than 100 nm will keep within the effective resolution of the indentation instruments). Again, about $N = 300$ tests and a grid size of $l = 5$ μm satisfies the final criteria.

The shale samples were tested for both target indentation depths, indenting in the (macroscopically identified) direction of symmetry ($x3$) and normal to the direction of symmetry ($x1$), over a range of loads and see how the responses varied for the different load cases. A few iterations demonstrated that a maximum indentation load of $P = 0.3$ mN on shale materials yielded an average maximum indentation depth of $h \approx 250$ nm, while a maximum indentation load of $P = 4.8$ mN gave maximum indentation depths in the range of $h \approx 1$ μm. For the pore throat radius giving $d = 10$ nm, and the larger indentation depths, Eq. (4) gives a reference length of the rev, $\mathscr{L}$, on the order of 100 nm. For the smaller depths, the reference length is smaller. Fig. 3 shows a 3-D representation of the surface of the sample after an indent. This test achieved the target indentation depth, and it is clear that the indentation depth is greater than the scale of the roughness and that the indentation response will depend on the composite behavior of a large number of individual clay particles.

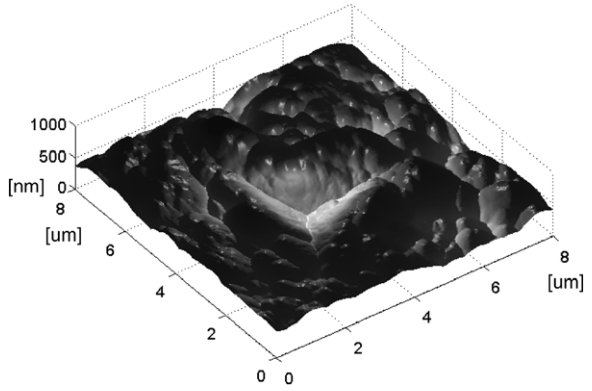


Fig. 3. Three-dimensional representation of AFM data, after an indent. The indent is clearly deeper than scale of the roughness, and is large enough to feel the response of a porous composite of clay particles.

A summary of the average achieved indentation loads, maximum indentation depths, grid size, and number of tests for each sample in each direction, is presented in Table 3.

3.1.1. Two dominant shapes for P–h curves

Fig. 4 displays three representative indentation curves to a maximum load of 4.8 mN, with the corresponding AFM images of the residual imprint of the indents. Two distinct types of indentation responses are evident. The response associated with indentation on quartz inclusions (curve (a) and AFM image (A) in Fig. 4) gives a lower indentation depth, a higher unloading slope (and higher indentation modulus) and exhibits only a small amount of irrecoverable deformation. The response associated with indentation in clay (curves (b) and (c) and AFM images (B) and (C) in Fig. 4) is different. These responses have deeper indentation depths,

Table 3
Indentation test parameters

Nom. load (mN)	Sample	Mean load, P (mN)	Mean depth, h (nm)	No. of tests, N
x1-direction				
0.3	Shale 1	0.324	854	401
	Shale 2	0.329	249	208
	Shale 3	0.282	112	400
	Shale 7	0.227	142	400
	Pierre shale	0.351	597	335
	Light shale	0.310	327	273
	Dark shale	0.268	181	300
	Low-clay shale	0.316	580	242
4.8	Shale 1	4.806	1810	388
	Shale 2	4.830	1189	336
	Shale 3	4.852	584	382
	Shale 7	4.776	678	400
	Pierre shale	4.801	1926	299
	Light shale	4.884	455	297
	Dark shale	4.778	624	299
	Low-clay shale	4.830	2204	224
x3-direction				
0.3	Shale 1	0.222	483	292
	Shale 2	0.308	183	292
	Shale 3	0.304	124	299
	Shale 7	0.295	145	296
	Pierre shale	0.327	517	259
	Light shale	0.287	81	299
	Dark shale	0.311	294	293
	Low-clay shale	0.323	650	242
4.8	Shale 1	4.741	1713	289
	Shale 2	4.956	692	428
	Shale 3	4.850	613	300
	Shale 7	4.865	689	291
	Pierre shale	4.850	2182	295
	Light shale	4.898	344	299
	Dark shale	4.854	824	294
	Low-clay shale	4.844	2551	253

lower unloading slopes (and lower indentation moduli) and exhibit a great deal of irrecoverable deformation. The indentation on porous clay in the $x1$-direction (curves and AFM images (b) in Fig. 4) gives a smaller indentation depth and steeper unloading slope than the indentation on porous clay in the $x3$-direction (curves and AFM images (c) in Fig. 4), which testifies to the anisotropic behavior of the porous clay phase in shales. There are possibilities for responses ranging between the characteristic clay and inclusion responses, and sometimes the distinction between responses is a bit unclear. The statistical deconvolution procedure gives results which are easier to obtain and interpret, with guidance from the individual responses providing a useful insight into the relationships between materials and indentation responses.

3.1.2. Deconvolution results

Fig. 5 displays a typical set of frequency plots derived from the indentation experiments and analysis on Shale 3 in the $x3$-direction. Two load cases are shown here; the top two plots in Fig. 5 are from the 0.3 mN tests, which have an average indentation depth of 124 nm, the bottom two plots are from the 4.8 mN tests, which have an average indentation depth of 613 nm. Similar plots are obtained for the other shales, in both indentation directions, $x1$ and $x3$. In each case, we observe two peaks (plus another peak associated with the quartz inclusions) at low indentation loads, corresponding to low indentation depths. For the deeper indentations, these two peaks converge, via mechanical homogenization, to one single peak associated with the porous clay (with the quartz remaining as before). Tables presented in the Appendix (Tables 4–7) summarize the results of the deconvolution procedure for all shales, for both load cases and both indentation directions.

3.2. Porous clay composite scaling –"deep indentations"

Consider first the porous clay composite sensed by the deeper indentation depths. We seek a simple relationship between the composition of the shale materials and the results of indentation measurements and deconvolution. Recall the clay packing density (10) as a volume fraction parameter. This parameter is the simplest possible volumetric simplification of mineralogy – it is a measure which depends only on overall porosity and on inclusion (non-clay) volume fraction. It is, however, an excellent starting point for examining overall, first-order trends.

The scaling of the indentation moduli (in both the $x1$- and $x3$-directions) and hardness with the clay packing density is quite interesting. It is shown graphically in Fig. 6. The x-intercept of these relationships ("going downward" along the trend) gives information about the micro(nano)-structure, while the y-intercept of this relationship ("going upward") gives material invariant stiffness and hardness properties. Possible explanations regarding the shale sample with the lowest clay packing density are considered in the context of this analysis.

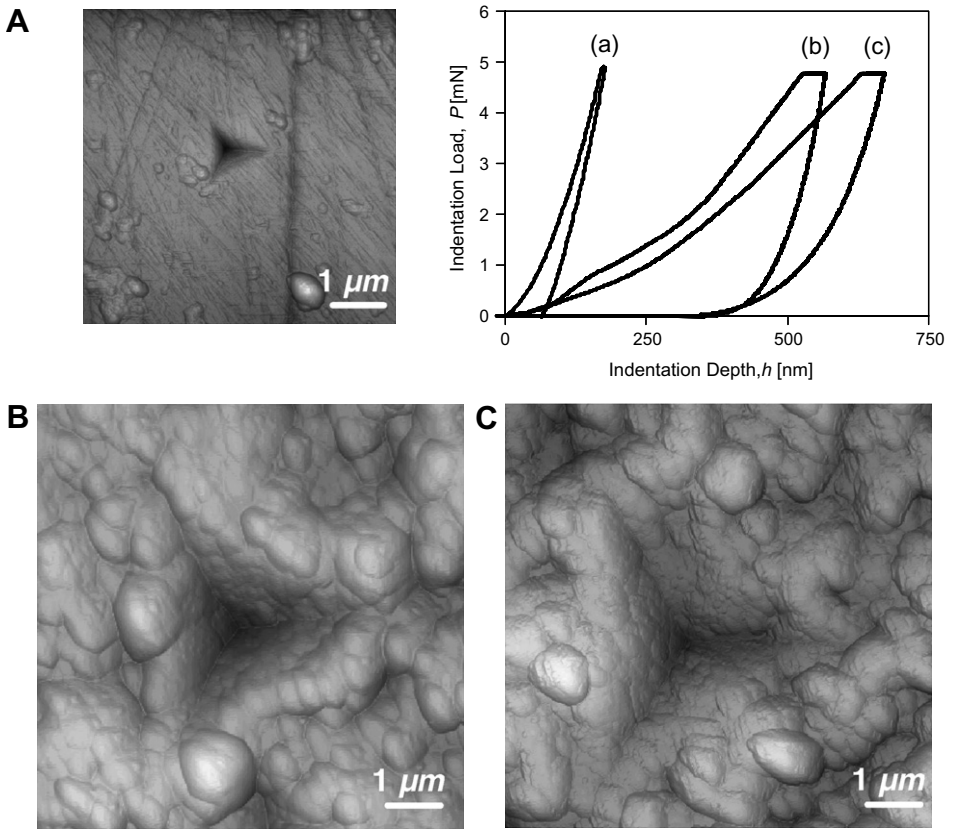


Fig. 4. Indentation responses on shales indented to a maximum load of 4.8 mN. Curve (a) is typical of indentation on a quartz inclusion, while the curves (b) and (c) are typical of indentation on the porous clay composite. Curve (b) is from indentation in the $x1$-direction and curve (c) is from indentation in the $x3$-direction. The other three images are AFM photo simulations after a 4.8 mN indent on (A) a quartz inclusion in the $x3$-direction, (B) the porous clay composite in Shale 3 in the $x1$-direction, and (C) the porous clay composite in Shale 3 in the $x3$-direction.

The relationship between the indentation hardness (in both the $x1$- and $x3$-directions) and the clay packing density is similar to the indentation modulus relationship, although there is one key difference. While the indentation modulus measurements are clearly anisotropic, the hardness measurements are clearly isotropic. This is shown graphically by comparison of the top and bottom graphs of Fig. 6. Although there is some scatter in the data, the vertical "error bars" (that represent the standard deviation of the peaks) in the hardness plot (Fig. 6, bottom) are consistently overlapping. There is no trend of one direction giving a larger hardness than another.

3.2.1. Mechanical microstructure

There is a trend of the indentation data to intercept a zero stiffness and zero hardness at a packing density of $\eta_0 \geqslant 0.5$. We interpret this intercept as a percolation threshold of the porous clay phase below which the porous clay in a shale material loses its strength and stiffness properties. This interpretation gives an important insight into the mechanical microstructure of the shale. This insight arises from percolation theory in combination with the polycrystal/self consistent mechanical model.

Indeed, the percolation threshold, the volume fraction of particles required to form a continuous force path through the system, changes with the particle shape. The packing of particles is a focus of attention in the granular physics community, and several researchers have attempted to understand limit packings of spheroids. Onoda and Liniger (1990) determined that the random-loose packing fraction of uniform spheres at the limit of zero gravitational force is 0.555 ± 0.005, which corresponds to a sphere packing at its rigidity-percolation threshold. Buchalter and Bradley (1994)

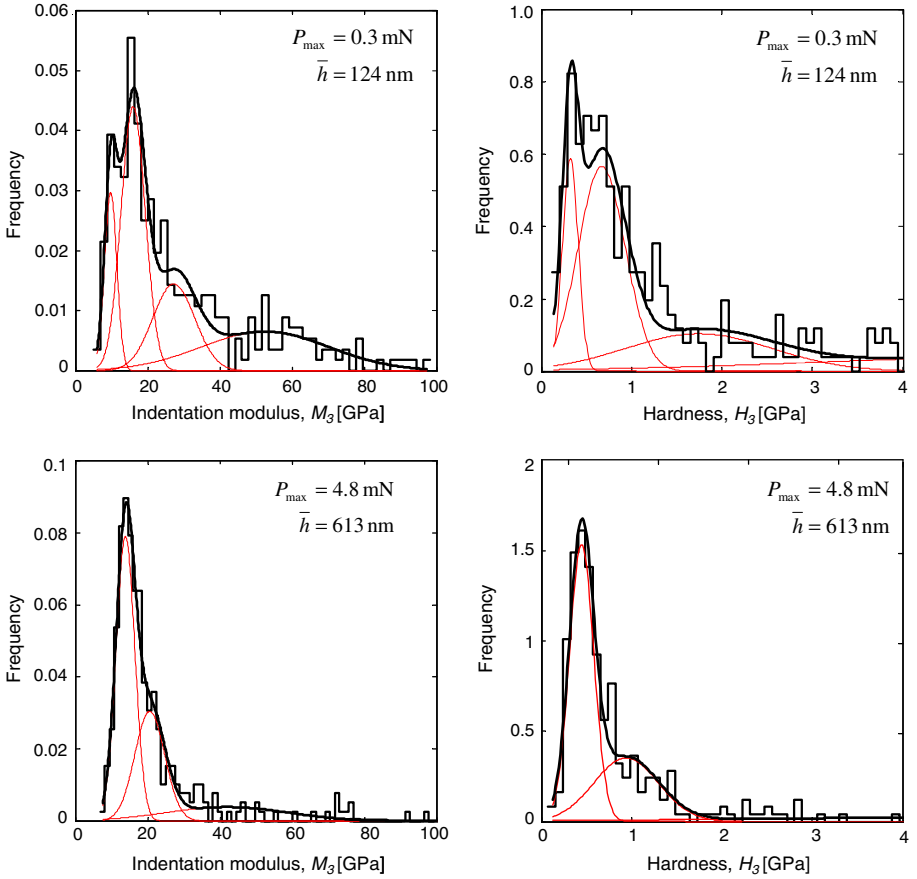


Fig. 5. PDFs of indentation modulus (left) and hardness (right) for Shale 3 in the $x3$-direction to a maximum indentation load of 0.3 mN (top) and 4.8 mN (bottom). Note the convergence of the two left-most peaks in the low-load tests (to) to one peak in the high-load tests (bottom).

performed Monte Carlo simulations of the pouring of oblate and prolate ellipsoids, and showed that the increase of asphericity yields to lowering the percolation threshold. Similar results have been obtained by Coehlo et al. (1997) using a sequential deposition algorithm of rigid particles of different shapes. The overall trend which thus emerges is that a pecolation threshold of $\eta_0 \geqslant 0.5$ is indicative of a negligible influence of the particle aspect ratio on the mechanical behavior.

The polycrystal, or self consistent model of micromechanics, which originated independently from Hershey (1954) and Kröner (1958), recognizes the percolation threshold associated with granular materials. The model treats each phase of the composite as a phase surrounded by the averaged composite, hence the self-consistent terminology. In this manner, the porous clay composite is treated as a disordered system, with each individual clay particle and pore space replaced by an abstraction of an averaged porous composite. Kröner's original work considered a perfectly random distribution of contact surfaces between particles. This averaged random contact may be represented, mechanically, by a sphere. For the spherical case, the percolation threshold of the polycrystal model is $\eta_0 = 0.5$. Non-spherical shapes introduce some order to the system, while a sphere itself is a "perfectly disordered" particle shape in the context of the self-consistent model (Zaoui, 2002).

A model of the porous clay morphology can be synthesized given that a percolation threshold of $\eta_0 \geqslant 0.5$ is observed in the indentation results. The mechanical morphology of the porous clay, at the nano-scale, is a pack of spherical objects; any other shape would give a lower percolation thresh-

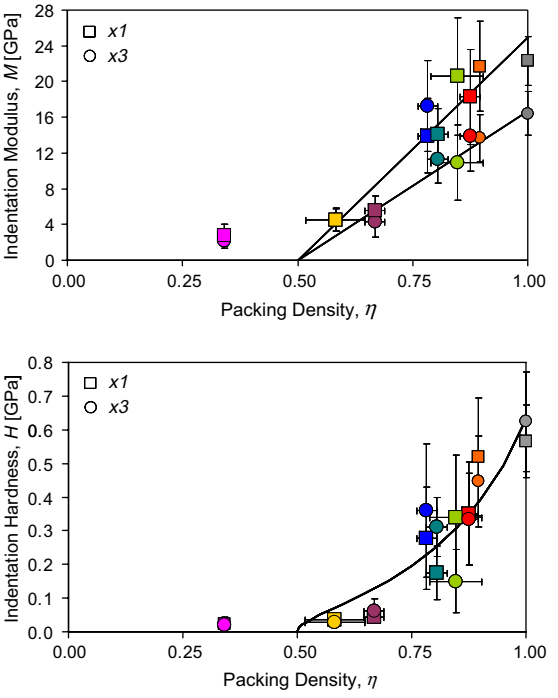


Fig. 6. Indentation modulus, M, (top) and hardness, H, (bottom) measured in both directions, versus clay packing density, η. The horizontal error bars in η represent the minimum and maximum packing densities determined by porosimetry or bulk density (see Table 1). The vertical error bars in M and H represent the standard deviation of the fitted peaks from the deconvolution analysis.

old. This observation of mechanical morphology is clearly distinct from the visible particle shape of clays as evidenced by microscopic imaging (Fig. 1). Despite their visible aspect ratio, the percolation threshold testifies to the randomness of the orientation of the contact surfaces between particles.

3.2.2. Material invariant properties

The material invariant properties are found as the asymptotic properties that a pack of clay particles would have if they could pack to a packing density of $\eta = 1$. By 'material invariant' we mean properties that do not depend neither on the mineralogy nor the porosity of the specific shale material. These material invariant properties are not necessarily the pure clay mineral properties, as they still encompass the contact interaction between particles, in combination with the pure mineral properties.

Following the trend of the indentation moduli data with packing density upwards, towards an asymptotic packing density of $\eta = 1$, allows an estimation of the asymptotic stiffness of the porous clay in shale. This extrapolation gives values of $\lim_{\eta=1} M_1 = 25$ GPa, and $\lim_{\eta=1} M_3 = 16$ GPa, as shown in Fig. 6 (top). These values clearly demonstrate that the elasticity of the porous clay is anisotropic.

Similarly, we follow the trend of the hardness data with packing density towards an asymptotic packing density of $\eta = 1$ to estimate the asymptotic hardness of the porous clay in shale. This extrapolation gives values of $\lim_{\eta=1} H_1 \approx \lim_{\eta=1} H_3 = 0.62$ GPa, as shown in Fig. 6 (bottom). Unlike the stiffness, the material invariant hardness is isotropic.

3.3. Measurement of material invariant properties – "shallow depths"

Thus far, in seeking information regarding the porous clay phase of the shale materials, we have considered only the "deep" indentations that sense the porous clay composite. We now turn to "shallow" indentations that should sense locally densely packed areas of porous clay. Inspection of the progression of frequency plots in Fig. 5, as previously discussed, reveals that mechanical homogenization takes place with increasing load and indentation depth. That is, for the shallow indentation depths, the frequency plots in the top of Fig. 5 show two lower-value peaks which converge to a single peak in the series of deeper indentations. But a question arises: if the smallest-valued peaks at higher depths correspond to the homogenized porous clay composite, what is the meaning of the two smallest peaks measured from lower load tests?

Returning again to the scale separability criteria (Eqs. (3)–(5)) and considering the indentation depths for the smallest load (less than 400 nm), one possible explanation is that roughness effects intervene and create additional scatter in the data, because R_q is on the order of 150–300 nm (see Table 2). This explanation would be more plausible, however, if there was a simpler trend – that is, a consistent shift up or down in measured properties, or a broadening of the peak, indicating greater uncertainty in the measurements. The roughness explanation does not address the consistent observation

of two distinct peaks in both the modulus and hardness measurements. Furthermore, the roughness measurement, which describes vertical deviations from a planar surface, does not quantify the horizontal spatial frequency of these deviations; roughness therefore may not be the most relevant quantity to ascertain the quality of 'shallow' indentation tests.

Recalling the scale separability criterion (Eq. (4)) in the context of the small indentation depths is helpful. We have achieved our target depth, which is not large enough to sense the overall porous clay composite, nor is small enough to sense the mechanical response of a single clay particle (with typical dimensions of 0.5–2 μm in the long direction). An inspection of the values from the "middle peak" of the low depth responses (the second distributions presented in Tables 6 and 7) along with a consideration of the relationship between clay packing density and mechanical properties however, reveals an extremely interesting result. Fig. 7 (top) displays the indentation moduli of the "middle peak" for each shale which was tested at average indentation depths below 400 nm. It is noteworthy that these values do not scale with the clay packing density, η, but instead have generally similar "middle peak" values in each direction. This similarity suggests that the tests measure the same property in each sample, despite the difference in mineralogy and porosity. The average values are $M_1 = 22.3 \pm 2.7$ GPa and $M_3 = 16.4 \pm 2.5$ GPa, with the uncertainty calculated as the standard error of the measured quantities. Fig. 7 (middle) displays the hardness of the "middle peak" for these shales. Again, there is no scaling with the clay packing density, but each shale has generally similar "middle peak" values, suggesting we measure the same property in each sample. The average values are $H_1 = 0.57 \pm 0.11$ GPa and $H_3 = 0.62 \pm 0.15$ GPa.

The surface fraction of "middle peak," which represents the probability of finding this response on a given sample surface, shows a very strong correlation with the overall clay packing density, as shown in Fig. 7 (bottom). This observation suggests that these measurements tested an asymptotic, locally densely packed state. Indeed, samples with a higher clay packing density have more solid phase; consequently, the likelihood of indenting on a locally densely packed state during a grid indentation campaign is greater in samples with higher clay packing density.

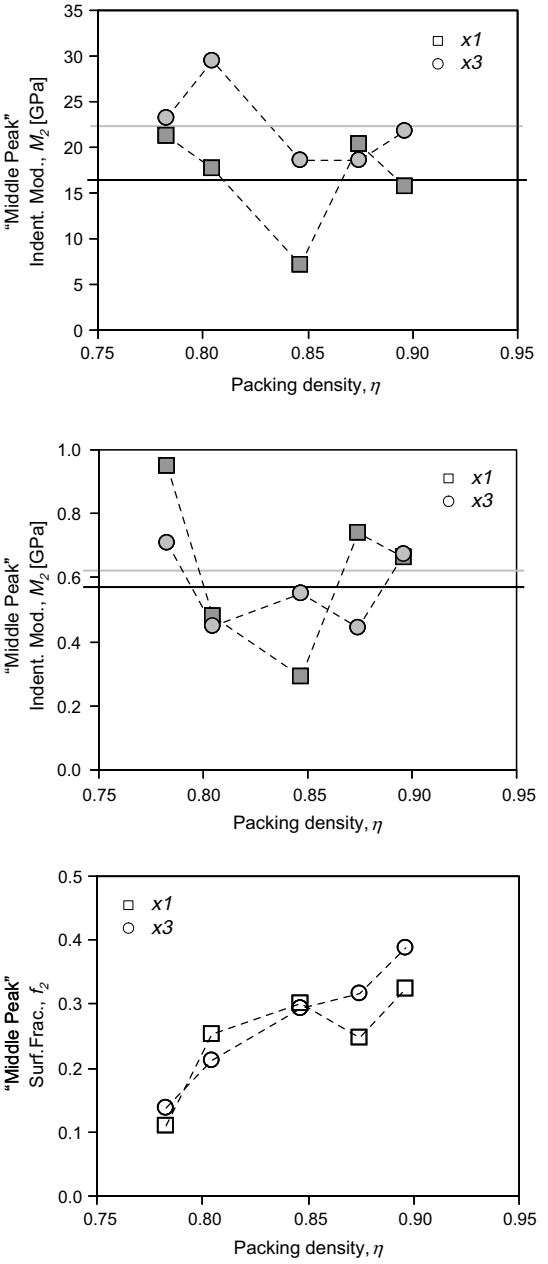


Fig. 7. "Middle peak" results versus clay packing density, η. Indentation moduli, M_2 in the $x1$- and $x3$-directions (top) and hardness, H_2, in the $x1$- and $x3$-directions (middle) show no consistent trend with clay packing density. The surface fraction of indents corresponding to the "middle peak," f_2, (bottom) increase with packing density.

Fig. 6 also shows the mean and standard error "middle peak" values, plotted at the asymptotic packing density of $\eta = 1$, along with the high-depth results discussed previously. The agreement of these

four points, at this asymptotic value, to the overall trends is remarkably good. This "middle peak" analysis presents another independent measurement of the material invariant properties of shale materials, the asymptotic stiffness and hardness of the porous clay phase. Moreover, this direct measurement by nanoindentation corresponds extremely well to the extrapolated measurement of the asymptotic stiffness and hardness, $\lim_{\eta=1} M_i$ and $\lim_{\eta=1} H_i$ ($i = 1,3$), of the homogenized porous clay phase presented above.

3.4. The low-clay shale

The low-clay shale presented difficulty in all aspects of this study, from polishing through analysis of the nanoindentation results. Its packing density, $\eta = 0.34$ (see Table 1) is apparently below the percolation threshold evident for the other tested shales, indicating that any porous clay present in the sample should have zero (or almost zero) stiffness. Experimentally, however, it is impossible to mechanically indent a material with zero stiffness. We suggest that the porous clay in this sample, agglomerated in local packings, has an extremely small stiffness when compared with the stiffness measured in the other samples. This suggestion is further supported by reported macroscopic dynamic measurements of the elasticity of this shale material, showing an almost isotropic macroscopic elasticity (namely, in Voigt notation, $C_{11}/C_{33} = 1.1$; $C_{66}/C_{44} = 1.2$; $C_{12}/C_{13} = 1.0$), that is close to the isotropic self-consistent estimate of the stiffness of a porous composite in which the silt inclusion (56%, see Table 1) solely contributes to the stiffness. In other words, this low-clay shale behaves rather like a sandstone than a shale having the porous clay phase as dominating load bearing phase. This observation and the careful inspection of the nanoindentation results implies that the indentation process may have compacted the clay particles in the low-clay shale, resulting in measurement of a local state just above the percolation threshold upon unloading (which is the data used in the analysis of each individual indentation experiment). This densification experienced in the low-clay shale indentation should be much less significant in the other samples. The other shale samples all have porous clay phases which are more densely packed to begin with. In fact, the other shales all have clay packing densities at or above the random close-packed limit (known as RCP[1]) of monosized spheres of $\eta = 0.64$ (Larsson et al., 1996). Consequently, we expect that any possible densification during indentation on such highly packed shales will have a second-order effect on the results of the indentation tests.

4. Discussion

4.1. Models

Modeling of the relationships between indentation moduli and clay packing density and indentation hardness and clay packing density help to give further insights to the mechanical properties of the porous clay in shale materials. To provide useful data, we wish to link indentation results to stiffness and strength parameters.

4.1.1. Indentation moduli

We choose a linear scaling between indentation moduli and clay packing density based on experimental observations. Considering the maximum percolation threshold of $\eta_0 = 0.5$ for a packing of spheres, a requirement of zero indentation modulus is enforced at this packing density, and we fit a straight line through the data (Fig. 6, top). The linear scaling is a simplistic choice, in keeping with the reductionist approach of this study.

In the isotropic case, the indentation modulus, M, reduces to the plane-stress elastic modulus:

$$M = \frac{E}{1-\nu^2} = 4G\frac{3K+G}{3K+4G}, \tag{11}$$

where E is the Young's modulus, ν is the Poisson's ratio, K is the bulk modulus, and G is the shear modulus. The experimental indentation moduli for the porous clay, however, are clearly anisotropic. Specifically, shale materials are expected to exhibit transversely isotropic elasticity, for which M_3 and M_1 are given by Delafargue and Ulm (2004):

[1] More recent concepts refer to the RCP as the maximally random jammed state (MRJ), corresponding to the least ordered among all jammed packings, which has been shown to have a density of 63.7%, and which is very close to the traditional definition of the random close-packed limit (see Donev et al., 2004), and references cited therein.

$$M_3 = 2\sqrt{\frac{C_{31}^2 - C_{13}^2}{C_{11}}\left(\frac{1}{C_{44}} + \frac{2}{C_{31}+C_{13}}\right)^{-1}}, \quad (12)$$

$$M_1 \approx \sqrt{\sqrt{\frac{C_{11}}{C_{33}}}\frac{C_{11}^2 - C_{12}^2}{C_{11}}M_3}, \quad (13)$$

where C_{ij} are components of the transversely isotropic stiffness tensor in Voigt notation. The indentation moduli can be considered a "snapshot" of the full elasticity tensor, and provide important, quantitative insights into the mechanical behavior of the porous clay in shale materials.

Direct measurements of clay minerals are scarce, but where they exist, they are distinctly different from the asymptotic values presented here. For example, classical values of the stiffness of muscovite give indentation moduli of $M_1 = 118$ GPa and $M_3 = 46$ GPa (Aleksandrov and Ryzhova, 1961), compared with values of $M_1(\eta = 1) = 25$ GPa and $M_3(\eta = 1) = 16$ GPa found as the asymptotic indentation moduli measured by *in situ* shale indentation. These asymptotic values are therefore best understood a combination of the stiffness of the clay particles and the averaged contact stiffnesses between particles.

Other researchers have sought an isotropic representation of elastic parameters for porosity-free clays, by extrapolating the elasticity response of clay mixtures to zero porosity. These values would correspond to our asymptotic moduli measured by indentation. Hornby et al. (1994) found values of $K = 22.9$ GPa and $G = 10.6$ GPa, while Berge and Berryman (1995) report values of $K = 21.4$ GPa and $G = 6.7$ GPa. These isotropic elasticity values, input in Eq. (11), give isotropic indentation moduli of $M = 30$ GPa and $M = 21$ GPa, respectively. These isotropic estimates fall within the range of anisotropic values found from indentation. The variation in these estimates may well be explained by consideration of anisotropic effects.

Furthermore, it is interesting to compare the results of nanoindentation to macroscopically observed data, derived from ultrasonic pulse velocity measurements (UPV) on some of the same shales studied by nanoindentation. These measurements were performed with frequencies in the megahertz range, such that they sense the full composite behavior of the shale samples. To compare our indentation data with these UPV measurements, we need to subtract from the measured values the effect of the silt inclusions. This is not an easy task. In a very first approximation, we subtract the inclusion by means of the Reuss bound:

$$(C_{ij}^{\mathrm{UPV}})^{-1} = S_{ij}^{\mathrm{UPV}} = (1 - f_{\mathrm{inc}})S_{ij} - f_{\mathrm{inc}}S_{ij}^{\mathrm{inc}}, \quad (14)$$

where C_{ij}^{UPV} is the measured dynamic (undrained) stiffness, $C_{ij}^{\mathrm{inc}} = (S_{ij}^{\mathrm{inc}})^{-1}$ is the stiffness of the inclusions that occupy a volume fraction f_{inc} and which is assumed to be isotropic (bulk modulus $k_{\mathrm{inc}} = 37.9$ GPa, shear modulus $g_{\mathrm{inc}} = 44.3$ GPa); and $C_{ij} = (S_{ij})^{-1}$ is the sought stiffness of the porous clay composite (Ulm and Abousleiman, 2006). This simple Reuss bound neglects any hydraulic stiffening effect of the composite response under undrained conditions of the porous material (see e.g. Coussy, 2004; Scott and Abousleiman, 2005) or specific micromechanical features that affect the porous composite response (see e.g. Dormieux et al., 2002; Dormieux et al., 2006). While no-doubt a rough estimate, our aim here is to compare this estimate of the composite stiffness response with the indentation results. We achieve this by substituting the porous clay composite stiffness C_{ij} determined from (14) into the indentation moduli relations for transverse isotropic materials, Eqs. (12) and (13). Since the Reuss estimate gives a lower bound in composite upscaling estimates, the porous clay stiffness derived from (14) should give an upper bound for downscaling estimates. Fig. 8 displays the match between indentation moduli estimated from UPV measurements with the indentation moduli measured di-

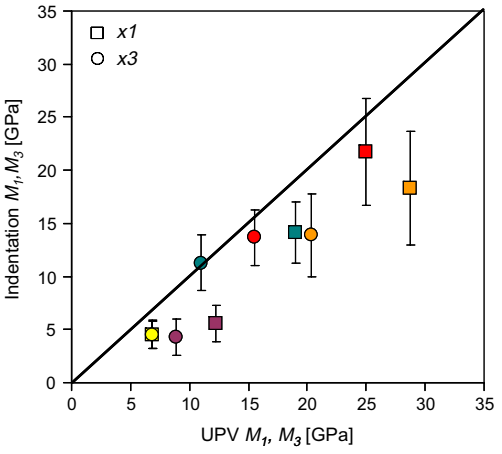


Fig. 8. Indentation Moduli of porous clay in both directions for Pierre Shale and Shales 1, 2, 3, and 7, obtained from indentation and from Reuss calculations of UPV measurements (UPV measurements on Pierre Shale from White et al. (1983)). The overall match is good, with the Reuss estimate from UPV giving an upper bound, as expected.

rectly for the same samples. Overall, the match is quite good, with the Reuss downscaling estimate giving an upper bound, as expected.

4.1.2. Hardness

The hardness measurement may be related to strength properties. Empirical relationships between hardness and strength properties have existed for some time, beginning with Brinell (1901) and continuing with Tabor (1948), who considered a rigid cohesive plastic solid indented by a frictionless rigid wedge. Our initial assumption, however, is that the shale material on which we indent is cohesive-frictional.

Given a porous material with solid particles obeying a cohesive-frictional strength criterion, Cariou et al. (2007) demonstrate that the relationship is of the form:

$$H = c_s \mathscr{F}(\alpha, \eta, \eta_0, \theta), \tag{15}$$

where c_s is the cohesion of the solid particle, and $\mathscr{F}(\alpha, \eta, \eta_0, \theta)$ is a dimensionless function of the friction coefficient, α, the packing density, η, and the percolation threshold, η_0, of the porous material, and the cone angle, θ, of the indenter. For the Berkovich indenter geometry and a polycrystal morphology with percolation threshold of $\eta_0 = 0.5$, the function is given by

$$\mathscr{F}(\alpha, \eta, \eta_0, \theta) = \mathscr{F}(\alpha = 0)[1 + (1 + \eta)\alpha - (d - e\eta)\alpha^2 - (f - g\eta)\alpha^5], \tag{16}$$

where $\mathscr{F}(\alpha = 0)$ is the frictionless portion of the function given by

$$\mathscr{F}(\alpha = 0) = \frac{12\eta\left((a - b\eta)\sqrt{(2\eta - 1)(2 + \eta)}\right)}{(1 - c\eta)(2 + \eta)} \tag{17}$$

and where $a = 0.19567$, $b = 0.03739$, $c = 0.77999$, $d = 20.3138$, $e = 31.5352$, $f = 52.1817$, and $g = 99.3465$ are all constants associated with the Berkovich indenter geometry and a polycrystal morphology with percolation threshold of $\eta_0 = 0.5$.

The model parameters, the friction coefficient, α, and the solid particle cohesion, c_s, were estimated by minimizing the quadratic error between the experimental hardness data and the model response. The optimized model results are the solid lines displayed in Fig. 6 (bottom), and a first inspection of the fits, which virtually overlap, confirms the assertion that the hardness behavior is isotropic. The estimated strength parameters of the porous clay, which form the basis for these fits, are also remarkable. The porous clay in shale is found to be frictionless (i.e. pressure insensitive) in strength, with cohesion forces providing the only resistance to permanent deformation. The solid particle cohesion is $c_s = 126$ MPa in the $x1$-direction and $c_s = 122$ MPa in the $x3$-direction. Optimizing for both directions at once, the solid particle cohesion is found as $c_s = 126$ MPa.

4.2. Elementary building block of nano-mechanical shale behavior

The results of the indentation experiments and analysis can be synthesized into a relatively simple understanding of nano-mechanical shale behavior: The elementary building block is transversely isotropic in stiffness, with $M_1 = 25$ GPa and $M_3 = 16$ GPa characterizing the solid's response to particle-to-particle contact forces in the deposition plane and normal to it. In first order, these stiffness values appear to be insensitive to the clay mineralogy (see Table 1). The anisotropy of the elementary building block is a consequence of the deposition history undergone by shales. By contrast, the orientation of the contact surfaces between particles possess a high degree of randomness, which is evidenced by a percolation threshold of $\eta_0 \geqslant 0.5$. Therefore, despite their plate- or sheet-like structures, the elementary clay particles behave at the nano-mechanical scale like spheres, with no privileged orientation of the particle-to-particle contact surfaces. This randomness of the contact surface orientation between particles translates into an isotropic and frictionless strength behavior of the elementary particle, having a cohesion of $c_s = 126$ MPa, which translates into a yield strength of $Y_s = 218$ MPa. In contrast to the elastic particle-to-particle contact, which activates intra-particle mechanical properties, the origin of this isotropic and frictionless strength behavior originates from local bonds at the particle-to-particle contact surface. Fig. 9 displays a cartoon illustrating the logical origin of the elementary building block for stiffness and for strength behavior.

This nano-mechanical building block of shales provides a suitable framework to revisit the origin of the macroscopically observed anisotropy in shale materials, which continues to be a topic of ongoing debate. Several researchers have attempted to explain the observation of increased anisotropy with increased density by suggesting that the plate-like clay minerals are highly anisotropic (Sayers, 1994,

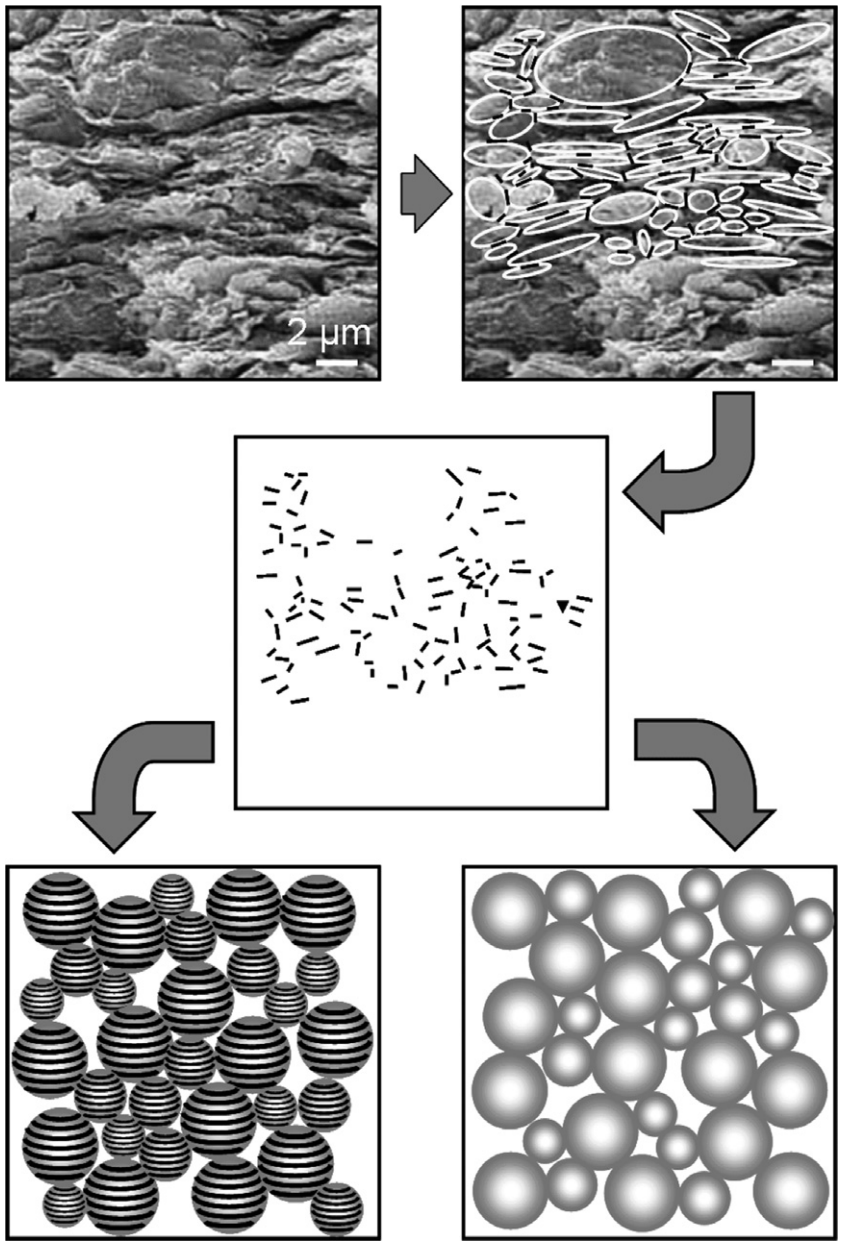


Fig. 9. The logical origin of the elementary building block for stiffness and strength behavior of porous clay in shale. The top left image is an ESEM image showing individual clay particles. The top right image has traces of particle shapes. The middle image shows the contact surfaces between traces of particle shapes. While orientation is clearly visible in the particle shape diagram, an ordered orientation of contact surface is much less apparent. As a result, the mechanical representation of this averaged random contact is a sphere. For stiffness (bottom left), the spherical building block is anisotropic, for strength (bottom right) it is isotropic.

Draege et al., 2006). This suggests that shales with more tightly packed clays are more anisotropic. Other researchers, beginning with Hornby et al. (1994), have attempted to apply effective-medium modeling to anisotropic solids by introducing anisotropic pore shapes. They then demonstrate that anisotropy decreases as porosity decreases. Their analysis assumes that an asymptotically densely packed state of shale is isotropic, which conflicts with the assumption that the clay minerals themselves are highly anisotropic. Hornby later provided some experimental evidence (Hornby, 1998) for

increased anisotropy with increasing porosity, but porosity changes in these data came from changes in the sample confining pressure rather than porosity differences between samples. A review of the data available in the literature, starting with Kaarsburg's first velocity measurements (Kaarsberg, 1959), demonstrate that anisotropy tends to increase with a decrease in porosity.

Our understanding of the elementary building block of shale behavior provides a new way of understanding the origin of anisotropy. The spherical nano-mechanical shape of the building block rules out particle shape (and consequently pore shape) as a source of anisotropy in shales. The scaling of elasticity with clay packing density directly reveals that shales with a higher clay packing density are more anisotropic than shales with lower clay packing densities. This scaling agrees with the traditional inverse relationship between anisotropy and porosity. The intrinsic anisotropy of the elementary building block follows, in spirit, the measured anisotropy of clay minerals. The building block is better understood, however, as a mechanical representation of the contact morphology between many individual clay minerals.

5. Conclusion

The key result of this investigation is that there exists a nano-mechanical morphology distinct from the visible structure of the porous clay phase in shales. We obtained this result from the analysis of a massive indentation campaign:

1. Appropriate indentation depths for accessing the properties of the porous clay in shale materials were carefully chosen based on consideration of scale separability criterion.
2. The results of a massive indentation campaign were analyzed to identify individual material phase indentation moduli and hardness in a wide variety of shale materials of different mineralogy and porosity.
3. Scaling of the deeper indentation results provides evidence of a percolation threshold at a clay packing density of $\eta_0 \geqslant 0.5$, leading to an understanding of the porous clay as a nano-granular material, with a spherical mechanical morphology.
4. Scaling of the deeper indentation results provides estimates of asymptotic indentation moduli and hardness, which – in first order – are insensitive to mineralogy. These values were confirmed by an analysis of the shallow indentation results. The results show that the stiffness of the elementary particle is anisotropic, while the strength is isotropic.
5. A comparison with previously reported data for shale elasticity shows good agreement with the indentation results. The *in situ* indentation provides a more detailed understanding of the elasticity of the porous clay in shales; the intrinsic anisotropy of the porous clay particle is captured by indentation.
6. An optimization of a hardness–strength relationship for a porous material with a cohesive frictional strength criterion demonstrates that the solid particles of porous clay have an isotropic and frictionless strength behavior.

These results lead to a morphological model for shale nano-mechanical behavior, characterized by an elementary building block with a spherical mechanical morphology. The building blocks exhibit nano-granular behavior, and their packing density is the controlling factor in the composite behavior of the porous clay. The elementary building block is anisotropic in elasticity, and this anisotropy is intrinsic rather than shape-related. The elementary building block is both frictionless and isotropic in strength. The model is apparently general to a wide variety of shale materials with varying mineralogies and porosities. It provides a fundamental understanding of the nature of shale behavior at this scale, and will permit future development of more sophisticated micromechanical models for both elasticity and strength.

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Appendix

See Tables 4–7

Table 4
Deconvolution results for indentation modulus from the 4.8 mN indentations

Sample	f_1	μ_1 (GPa)	σ_1 (GPa)	f_2	μ_2 (GPa)	σ_2 (GPa)	f_3	μ_3 (GPa)	σ_3 (GPa)
x1-direction									
Shale 1	0.52	5.54	1.69	0.34	10.57	3.33	0.14	22.97	9.07
Shale 2	0.61	12.25	4.81	0.26	22.15	5.10	0.14	53.40	26.15
Shale 3	0.59	21.67	5.01	0.27	32.39	5.70	0.14	63.35	20.45
Shale 7	0.62	18.28	5.36	0.21	28.81	5.17	0.18	59.45	21.49
Pierre shale	0.47	4.50	1.24	0.35	7.91	2.18	0.17	15.55	5.46
Light shale	0.17	13.95	4.16	0.19	31.92	9.84	0.64	71.72	27.18
Dark shale	0.52	20.57	6.57	0.24	35.58	8.43	0.23	69.53	20.06
Low-clay shale	0.63	2.82	1.24	0.22	9.71	5.23	0.15	36.99	17.58
x3-direction									
Shale 1	0.72	4.65	1.93	0.26	9.68	2.53	0.02	45.79	33.23
Shale 2	0.54	11.77	3.17	0.29	23.36	6.75	0.17	60.75	23.16
Shale 3	0.52	13.67	2.62	0.32	20.43	4.14	0.16	41.95	16.80
Shale 7	0.53	13.87	3.87	0.24	25.52	6.79	0.22	66.04	33.73
Pierre shale	0.49	4.74	1.45	0.42	8.41	2.23	0.09	23.02	5.67
Light shale	0.22	17.42	6.81	0.20	48.39	12.71	0.57	97.80	18.89
Dark shale	0.39	10.95	4.23	0.32	24.50	9.20	0.29	78.12	24.32
Low-clay shale	0.45	2.21	0.84	0.33	4.72	1.55	0.21	11.98	5.71

f_i, μ_i, and σ_i are the surface fraction, mean value, and standard deviation, respectively, of each phase for $i = 1$–3.

Table 5
Deconvolution results for hardness from the 4.8 mN indentations

Sample	f_1	μ_1 (GPa)	σ_1 (GPa)	f_2	μ_2 (GPa)	σ_2 (GPa)	f_3	μ_3 (GPa)	σ_3 (GPa)
x1-direction									
Shale 1	0.52	0.04	0.02	0.34	0.11	0.04	0.14	0.63	0.33
Shale 2	0.61	0.13	0.09	0.26	0.54	0.23	0.14	4.58	2.30
Shale 3	0.59	0.52	0.18	0.27	1.23	0.50	0.14	7.34	3.78
Shale 7	0.62	0.35	0.15	0.21	1.04	0.49	0.18	5.55	3.16
Pierre shale	0.47	0.04	0.01	0.35	0.10	0.04	0.17	0.49	0.26
Light shale	0.17	0.28	0.15	0.19	1.28	0.77	0.64	7.91	4.12
Dark shale	0.52	0.34	0.19	0.24	1.61	0.75	0.23	7.82	4.11
Low-clay shale	0.63	0.03	0.01	0.22	0.16	0.08	0.15	3.25	1.71
x3-direction									
Shale 1	0.72	0.07	0.04	0.26	0.25	0.11	0.02	0.51	0.11
Shale 2	0.54	0.34	0.11	0.29	0.84	0.44	0.17	5.56	3.02
Shale 3	0.52	0.45	0.14	0.32	0.94	0.36	0.16	5.04	2.72
Shale 7	0.53	0.33	0.14	0.24	0.84	0.37	0.22	7.02	4.10
Pierre shale	0.49	0.03	0.01	0.42	0.07	0.03	0.09	3.71	1.01
Light shale	0.22	0.56	0.40	0.20	4.49	2.72	0.57	12.81	2.63
Dark shale	0.39	0.15	0.09	0.32	0.75	0.45	0.29	10.25	4.38
Low-clay shale	0.45	0.02	0.01	0.33	0.06	0.03	0.21	0.33	0.18

f_i, μ_i, and σ_i are the surface fraction, mean value, and standard deviation, respectively, of each phase for $i = 1$–3.

Table 6
Deconvolution results for indentation modulus from the 0.3 mN indentations

Sample	f_1	μ_1 (GPa)	σ_1 (GPa)	f_2	μ_2 (GPa)	σ_2 (GPa)	f_3	μ_3 (GPa)	σ_3 (GPa)	f_4	μ_4 (GPa)	σ_4 (GPa)
x1-direction												
Shale 1	0.49	2.41	1.00	0.20	4.50	1.09	0.17	13.17	5.45	0.16	40.70	21.45
Shale 2	0.21	6.63	3.50	0.38	15.14	5.00	0.23	29.54	8.01	0.18	69.00	31.11
Shale 3	0.20	12.78	2.90	0.32	21.72	4.62	0.22	33.60	7.26	0.26	64.62	23.18
Shale 7	0.27	11.46	3.48	0.27	18.31	2.92	0.15	28.87	5.01	0.31	66.91	22.42
Pierre shale	0.27	1.59	0.59	0.39	3.38	1.20	0.29	9.30	4.53	0.05	46.30	7.34
Light shale	0.15	13.10	3.60	0.13	23.84	7.14	0.11	38.95	7.67	0.61	72.97	26.04
Dark shale	0.38	6.56	3.52	0.30	18.63	6.93	0.32	68.53	26.26	–	–	–
Low-clay shale	0.34	1.35	0.57	0.25	3.12	1.19	0.20	9.41	3.40	0.21	29.91	13.84
x3-direction												
Shale 1	0.23	1.08	0.55	0.34	3.00	1.37	0.25	7.93	3.56	0.18	45.12	31.95
Shale 2	0.35	10.53	3.05	0.24	17.20	3.02	0.19	28.53	8.30	0.22	66.17	26.52
Shale 3	0.13	9.59	1.75	0.37	15.84	3.35	0.21	27.16	5.85	0.29	52.68	17.64
Shale 7	0.24	10.38	3.79	0.32	20.32	5.70	0.24	54.59	16.09	0.20	102.73	32.05
Pierre shale	0.40	3.07	1.13	0.28	5.45	0.88	0.19	9.32	2.92	0.13	29.66	14.67
Light shale	0.10	6.43	3.67	0.14	21.05	9.48	0.12	49.27	18.75	0.65	95.09	19.29
Dark shale	0.22	2.16	1.27	0.29	7.21	3.78	0.40	34.43	7.08	0.08	77.16	14.18
Low-clay shale	0.34	1.25	0.48	0.23	3.51	1.54	0.14	10.13	4.82	0.29	81.78	45.05

f_i, μ_i, and σ_i are the surface fraction, mean value, and standard deviation, respectively, of each phase for $i = 1$–4.

Table 7
Deconvolution results for hardness from the 0.3 mN indentations

Sample	f_1	μ_1 (GPa)	σ_1 (GPa)	f_2	μ_2 (GPa)	σ_2 (GPa)	f_3	μ_3 (GPa)	σ_3 (GPa)	f_4	μ_4 (GPa)	σ_4 (GPa)
x1-direction												
Shale 1	0.49	0.028	0.015	0.20	0.07	0.02	0.17	0.24	0.12	0.16	4.08	2.35
Shale 2	0.21	0.052	0.03	0.38	0.17	0.08	0.23	0.47	0.17	0.18	3.44	1.57
Shale 3	0.20	0.29	0.10	0.32	0.67	0.26	0.22	2.66	1.36	0.26	8.22	4.20
Shale 7	0.27	0.21	0.08	0.27	0.43	0.14	0.15	1.35	0.77	0.31	7.43	4.52
Pierre shale	0.27	0.019	0.006	0.39	0.05	0.02	0.29	0.14	0.08	0.05	10.19	0.14
Light shale	0.15	0.20	0.11	0.13	0.72	0.29	0.11	2.09	0.78	0.61	8.10	3.96
Dark shale	0.38	0.15	0.08	0.30	0.56	0.31	0.32	9.53	5.72	–	–	–
Low-clay shale	0.34	0.021	0.009	0.25	0.05	0.02	0.20	0.17	0.10	0.21	1.61	1.10
x3-direction												
Shale 1	0.23	0.010	0.007	0.34	0.046	0.024	0.25	0.23	0.12	0.18	3.47	1.93
Shale 2	0.35	0.21	0.09	0.24	0.45	0.15	0.19	1.04	0.43	0.22	4.02	2.55
Shale 3	0.13	0.32	0.09	0.37	0.67	0.26	0.21	1.75	0.82	0.29	5.91	2.69
Shale 7	0.24	0.22	0.13	0.32	0.74	0.38	0.24	3.75	2.02	0.20	10.33	2.52
Pierre shale	0.40	0.029	0.013	0.28	0.063	0.02	0.19	0.18	0.1	0.13	1.24	0.79
Light shale	0.10	0.16	0.13	0.14	0.93	0.53	0.12	5.50	1.92	0.65	15.78	5.05
Dark shale	0.22	0.04	0.03	0.29	0.29	0.19	0.40	3.44	1.67	0.08	7.32	2.21
Low-clay shale	0.34	0.012	0.007	0.23	0.053	0.029	0.14	0.28	0.14	0.29	7.96	4.59

f_i, μ_i, and σ_i are the surface fraction, mean value, and standard deviation, respectively, of each phase for $i = 1$–4.

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